

untangle::actuator

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1 actuator	1
2 Topic Index	3
2.1 Topics	3
3 Namespace Index	5
3.1 Namespace List	5
4 Class Index	7
4.1 Class List	7
5 File Index	9
5.1 File List	9
6 Topic Documentation	11
6.1 namespace untangle: functions	11
6.1.1 Detailed Description	11
6.1.2 Function Documentation	11
6.1.2.1 connect()	11
6.1.2.2 bind() [1/2]	12
6.1.2.3 bind() [2/2]	13
7 Namespace Documentation	15
7.1 untangle Namespace Reference	15
7.1.1 Detailed Description	15
7.1.2 Function Documentation	15
7.1.2.1 last_arg()	15
8 Class Documentation	17
8.1 untangle::actuator< action_t > Struct Template Reference	17
8.1.1 Detailed Description	18
8.1.2 Member Typedef Documentation	18
8.1.2.1 actions_t	18
8.1.2.2 results_t	18
8.1.3 Member Function Documentation	19
8.1.3.1 operator=()	19
8.1.3.2 type()	19
8.1.3.3 reset()	19
8.1.3.4 operator()()	19
8.1.3.5 invoke_action()	20
8.1.3.6 add() [1/2]	20
8.1.3.7 add() [2/2]	20
8.1.3.8 remove() [1/2]	21
8.1.3.9 remove() [2/2]	21
8.1.3.10 is_connected()	22
8.1.3.11 has_action()	22

8.2 untangle::invalid_action Struct Reference	22
8.2.1 Detailed Description	23
8.2.2 Constructor & Destructor Documentation	23
8.2.2.1 invalid_action()	23
8.2.3 Member Function Documentation	23
8.2.3.1 what()	23
9 File Documentation	25
9.1 actuator.hpp	25
Index	29

Chapter 1

actuator

An actuator is a generic callable that may invoke a list of actions(callables of type `std::function<...>`).

Its goal is to provide a light and simple mechanism to connect components that want to communicate with each other. It works similar as the "signal and slots" mechanisms from other frameworks, but much simplified.

Modern programming languages do not offer class intrinsic interfaces that would improve significantly how objects are connected and how they communicate. Instead, languages, like C++, are using external interfaces, that are not only cluttering the code, but they don't feel "natural" as they are opposed to what an interface is in reality. Basically the objects and interfaces are not separated, and the interface is intrinsic to the object(think on the cogs in a mechanism). A generic callable is an attempt to mitigate this problem. `async` implementation is such use case.

An actuator provides the results of all actions in a vector, in the same order as the actions were added.

Example: how to use actuator instead of polymorphism

```
void rotate_shapes(const std::vector<shape*>& shapes, int angle)
{
    for (const auto& s : shapes)
    {
        s->rotate(angle);
    }
}

std::shared_ptr<triangle> t(new triangle);
std::shared_ptr<circle> c(new circle);
std::shared_ptr<square> s(new square);

// using polymorphism
std::vector<shape*> shapes;
shapes.push_back(t.get());
shapes.push_back(c.get());
shapes.push_back(s.get());

rotate_shapes(shapes, 10);

// using actuator
auto action1 = untangle::bind(t, &triangle::rotate);
auto action2 = untangle::bind(c, &circle::rotate);
auto action3 = untangle::bind(s, &square::rotate);
auto actuator_rotate = untangle::connect(action1, action2, action3);
actuator_rotate(20);
```

For convenience there are provided helpers methods to "connect" to an initial list of "actions", or to create bindings to class methods.

Please check the manual in [doc/refman.pdf](#) for further references.

Chapter 2

Topic Index

2.1 Topics

Here is a list of all topics with brief descriptions:

namespace untangle: functions [11](#)

Chapter 3

Namespace Index

3.1 Namespace List

Here is a list of all documented namespaces with brief descriptions:

untangle	Interface to untangle::actuator functor	15
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Chapter 4

Class Index

4.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

untangle::actuator< action_t >	
An actuator is a functor that can trigger a dynamic list of actions (of type std::↵ function<...>)	17
untangle::invalid_action	
Invalid action exception	22

Chapter 5

File Index

5.1 File List

Here is a list of all documented files with brief descriptions:

actuator.hpp	25
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Chapter 6

Topic Documentation

6.1 namespace untangle: functions

Functions

- `template<typename action_t, typename... Actions>`
`actuator< action_t > untangle::connect (action_t &A1, Actions &... An)`
Creates an actuator holding an initial list of actions.
- `template<typename class_t, typename T, typename action_t = std::function<typename function_remove_const<↵`
`T>::type>>`
`action_t untangle::bind (const std::shared_ptr< class_t > &obj, T class_t::*method)`
Binding to a class function member.
- `template<typename class_t, typename T, typename action_t = std::function<typename function_remove_const<↵`
`T>::type>>`
`action_t untangle::bind (class_t *obj, T class_t::*method)`
Binding to a class method.

6.1.1 Detailed Description

6.1.2 Function Documentation

6.1.2.1 connect()

```
template<typename action_t, typename... Actions>
actuator< action_t > untangle::connect (
    action_t & A1,
    Actions &... An)
```

Creates an actuator holding an initial list of actions.

Parameters

A1	- The first action. It is specified as <code>std::function<...></code> .
----	--

An	- Any number of further actions, of the same type as A1.
----	--

Returns

An [actuator](#).

Example:

```
void rotate_shapes(const std::vector<shape*>& shapes, int angle) {
    for (const auto& s : shapes) {
        s->rotate(angle);
    }
}

auto action1 = untangle::bind(t, &triangle_mock::rotate);
auto action2 = untangle::bind(c, &circle_mock::rotate);
auto action3 = untangle::bind(s, &square_mock::rotate);

auto actuator_rotate = untangle::connect(action1, action2, action3);
actuator_rotate(20);
```

6.1.2.2 bind() [1/2]

```
template<typename class_t, typename T, typename action_t = std::function<typename function_remove_const<T>::<
type>>>
action_t untangle::bind (
    const std::shared_ptr< class_t > & obj,
    T class_t::* method)
```

Binding to a class function member.

It returns a `std::function(lambda)` that wraps the function member. It may be used to provide an action for [connect\(\)](#) or [actuator::add\(\)](#).

Remarks

It requires a shared pointer to the class type. A `std::weak_ptr` to it is captured internally in a lambda, so the binding can always check whether the shared object is still alive without keeping it alive itself. Therefore, it is safe to use actions provided by this binding inside an [actuator](#), and safe for the action to outlive the caller's shared pointer.

Parameters

obj	- Class object.
method	- Pointer to function member. It is specified as <code>&<class type>::<function member></code>

Returns

`action_t` - A `std::function` that wraps the pointer to function member.

Remarks

If the class object gets invalid, invoking this binding will throw an exception of type [invalid_action](#).

6.1.2.3 bind() [2/2]

```
template<typename class_t, typename T, typename action_t = std::function<typename function_remove_const<T>::↵  
type>>  
action_t untangle::bind (  
    class_t * obj,  
    T class_t::* method)
```

Binding to a class method.

Attention

It is not safe to use this binding when the pointed-to object may be destroyed. A null pointer is detected and reported as a dead action, but a pointer to an already destroyed object can not be distinguished from a valid one. Prefer the overload taking a `std::shared_ptr`.

Remarks

It is provided for convenience of use: within a class it is safe to create bindings through this pointer. Binding a null pointer produces an action that is always dead: invoking it throws [invalid_action](#), and an [actuator](#) drops it.

Parameters

obj	- Pointer to class.
method	- Pointer to function member. It is specified as <code>&<class type>::<function member></code>

Returns

action_t - A `std::function` that wraps the pointer to function member.

Chapter 7

Namespace Documentation

7.1 untangle Namespace Reference

Interface to [untangle::actuator](#) functor.

Classes

- struct [invalid_action](#)
Invalid action exception.
- struct [actuator](#)
An actuator is a functor that can trigger a dynamic list of actions (of type `std::function<...>`).

Functions

- `template<typename... Args>`
`auto last_arg (Args &&... args)`
Get last argument of params pack;.
- `template<typename action_t, typename... Actions>`
`actuator< action_t > connect (action_t &A1, Actions &... An)`
Creates an actuator holding an initial list of actions.
- `template<typename class_t, typename T, typename action_t = std::function<typename function_remove_const<↵`
`T>::type>>`
`action_t bind (const std::shared_ptr< class_t > &obj, T class_t::*method)`
Binding to a class function member.
- `template<typename class_t, typename T, typename action_t = std::function<typename function_remove_const<↵`
`T>::type>>`
`action_t bind (class_t *obj, T class_t::*method)`
Binding to a class method.

7.1.1 Detailed Description

Interface to [untangle::actuator](#) functor.

7.1.2 Function Documentation

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7.1.2.1 last_arg()

Template Parameters

Args	params template types
------	-----------------------

Parameters

args	params
------	--------

Returns

last param value in case it exist, otherwise 0.

Chapter 8

Class Documentation

8.1 untangle::actuator< action_t > Struct Template Reference

An actuator is a functor that can trigger a dynamic list of actions (of type `std::function<...>`).

```
#include <actuator.hpp>
```

Public Types

- using `actions_t` = `std::list<action_t*>`
Actions container type.
- using `results_t` = `std::vector<typename result_t::type>`
Results container type.

Public Member Functions

- `actuator & operator=` (const `actuator &other`)=default
Assignment operator.
- `action_t type` () const
Helper method to access the action type of this object.
- void `reset` ()
Remove all actions and any stored results.
- `template<typename... Args>`
void `operator()` (Args &&... args)
The call operator.
- `template<typename... Args>`
void `invoke_action` (const `std::string &name`, Args &&... args)
Invokes one single action associated with a key.
- void `add` (`action_t *action`)
Add an action to the actions list.
- void `add` (const `std::string &name`, `action_t *action`)
Add action to the actions map associated with a name.
- void `remove` (const `action_t *action`)
Remove an action from the actions list.
- void `remove` (const `std::string &name`)
Remove an action from actions map.
- bool `is_connected` () const
Check if this actuator is "connected" with other actions.
- bool `has_action` (const `std::string &name`) const
Check if there is certain named action.

Public Attributes

- [actions_t](#) actions
Actions list.
- [actions_map_t](#) [actions_map](#)
Named actions map.
- [results_t](#) results
Actions return values list.

8.1.1 Detailed Description

```
template<typename action_t>
struct untangle::actuator< action_t >
```

An actuator is a functor that can trigger a dynamic list of actions (of type `std::function<...>`).

Remarks

An actuator object can be constructed with an initial list of actions by [connect\(\)](#).

Template Parameters

<code>action_t</code>	Action type. It is specified as <code>std::function<...></code> .
-----------------------	---

8.1.2 Member Typedef Documentation

8.1.2.1 `actions_t`

```
template<typename action_t>
using untangle::actuator< action_t >::actions_t = std::list<action_t*>
```

Actions container type.

Remarks

The elements stored are of pointer type, that is required to implement the [remove\(\)](#) operation. `std::function` supports only equality operator for nullptr (two `std::function(s)` can not compare).

8.1.2.2 `results_t`

```
template<typename action_t>
using untangle::actuator< action_t >::results_t = std::vector<typename result_t::type>
```

Results container type.

It holds the return values of the actions that have a non-void return type. Upon the actuator invocation, the returns can be extracted from [results](#).

8.1.3 Member Function Documentation

8.1.3.1 operator=()

```
template<typename action_t>
actuator & untangle::actuator< action_t >::operator= (
    const actuator< action_t > & other) [default]
```

Assignment operator.

Example:

```
const auto t = std::make_shared<triangle_mock>();
const auto c = std::make_shared<circle_mock>();
const auto s = std::make_shared<square_mock>();

EXPECT_CALL(*t, rotate(testing::_)).WillOnce(testing::Return());
EXPECT_CALL(*c, rotate(testing::_)).WillOnce(testing::Return());
EXPECT_CALL(*s, rotate(testing::_)).WillOnce(testing::Return());
auto action1 = untangle::bind(t, &triangle_mock::rotate);
auto action2 = untangle::bind(c, &circle_mock::rotate);
auto action3 = untangle::bind(s, &square_mock::rotate);

auto actuator_rotate = untangle::connect(action1, action2, action3);
untangle::actuator<decltype(actuator_rotate.type())> actuator_rotate_1 = actuator_rotate;
actuator_rotate_1(20);

testing::Mock::VerifyAndClearExpectations(t.get());
testing::Mock::VerifyAndClearExpectations(c.get());
testing::Mock::VerifyAndClearExpectations(s.get());
```

8.1.3.2 type()

```
template<typename action_t>
action_t untangle::actuator< action_t >::type () const [inline]
```

Helper method to access the action type of this object.

Note

Intended to be used in expressions by `decltype(<actuator instance>.type())`

Returns

action_t

8.1.3.3 reset()

```
template<typename action_t>
void untangle::actuator< action_t >::reset () [inline]
```

Remove all actions and any stored results.

After this call the actuator is empty: `actuator::is_connected()` returns false.

8.1.3.4 operator()()

```
template<typename action_t>
template<typename... Args>
void untangle::actuator< action_t >::operator() (
    Args &&... args) [inline]
```

Parameters

args	- Arguments list must match the action arity.
------	---

8.1.3.5 invoke_action()

```
template<typename action_t>
template<typename... Args>
void untangle::actuator< action_t >::invoke_action (
    const std::string & name,
    Args &&... args) [inline]
```

Invokes one single action associated with a key.

Parameters

name	- Key associated with the action.
args	- Arguments list must match the action arity.

8.1.3.6 add() [1/2]

```
template<typename action_t>
void untangle::actuator< action_t >::add (
    action_t * action) [inline]
```

Add an action to the actions list.

Parameters

action	- Action to be added.
--------	-----------------------

Example:

```
const auto t = std::make_shared<triangle_mock>();
const auto c = std::make_shared<circle_mock>();
const auto s = std::make_shared<square_mock>();

EXPECT_CALL(*t, rotate(testing::_)).Times(2).WillRepeatedly(testing::Return());
EXPECT_CALL(*c, rotate(testing::_)).WillOnce(testing::Return());
EXPECT_CALL(*s, rotate(testing::_)).WillOnce(testing::Return());
auto action1 = untangle::bind(t, &triangle_mock::rotate);
auto action2 = untangle::bind(c, &circle_mock::rotate);
auto action3 = untangle::bind(s, &square_mock::rotate);

auto actuator_rotate = untangle::connect(action1, action2, action3);
actuator_rotate.add(&action1);
actuator_rotate(20);

testing::Mock::VerifyAndClearExpectations(t.get());
testing::Mock::VerifyAndClearExpectations(c.get());
testing::Mock::VerifyAndClearExpectations(s.get());
```

8.1.3.7 add() [2/2]

```
template<typename action_t>
void untangle::actuator< action_t >::add (
```


Parameters

name	- Name of the action.
action	- Action to be added.

8.1.3.8 remove() [1/2]

```
template<typename action_t>
void untangle::actuator< action_t >::remove (
    const action_t * action) [inline]
```

Remove an action from the actions list.

An invalid action (empty std::function) is implicitly removed when `operator()()` is invoked.

Parameters

action	- Action to be removed.
--------	-------------------------

Example:

```
const auto t = std::make_shared<triangle_mock>();
const auto c = std::make_shared<circle_mock>();
const auto s = std::make_shared<square_mock>();

EXPECT_CALL(*t, rotate(testing::_)).Times(0);
EXPECT_CALL(*c, rotate(testing::_)).WillOnce(testing::Return());
EXPECT_CALL(*s, rotate(testing::_)).WillOnce(testing::Return());
auto action1 = untangle::bind(t, &triangle_mock::rotate);
auto action2 = untangle::bind(c, &circle_mock::rotate);
auto action3 = untangle::bind(s, &square_mock::rotate);

auto actuator_rotate = untangle::connect(action1, action2, action3);
actuator_rotate.remove(&action1);
actuator_rotate(50);

testing::Mock::VerifyAndClearExpectations(t.get());
testing::Mock::VerifyAndClearExpectations(c.get());
testing::Mock::VerifyAndClearExpectations(s.get());
```

8.1.3.9 remove() [2/2]

```
template<typename action_t>
void untangle::actuator< action_t >::remove (
    const std::string & name) [inline]
```

Remove an action from actions map.

Parameters

name	- Name of the action to remove.
------	---------------------------------

8.1.3.10 is_connected()

```
template<typename action_t>
bool untangle::actuator< action_t >::is_connected () const [inline]
```

Check if this actuator is "connected" with other actions.

Returns

- true - if the `actuator::actions` list is not empty.
- false - if the `actuator::actions` list is empty.

8.1.3.11 has_action()

```
template<typename action_t>
bool untangle::actuator< action_t >::has_action (
    const std::string & name) const [inline]
```

Check if there is certain named action.

Parameters

name	- Name associated with the action.
------	------------------------------------

Returns

- true - if name can be found in `actuator::actions_map`
- false - if name can not be found in `actuator::actions_map`

The documentation for this struct was generated from the following file:

- `actuator.hpp`

8.2 untangle::invalid_action Struct Reference

Invalid action exception.

```
#include <actuator.hpp>
```

Public Member Functions

- `invalid_action` (std::string text)
Construct a new invalid action object.
- `const char * what () const` noexcept override
The message text describing the reason of this exception.

8.2.1 Detailed Description

Invalid action exception.

Remarks

An action may be provided as a binding to a class function member, by using [untangle::bind\(\)](#). When the class object gets invalid, invoking the action will raise an exception to this type.

8.2.2 Constructor & Destructor Documentation

8.2.2.1 invalid_action()

```
untangle::invalid_action::invalid_action (  
    std::string text) [inline], [explicit]
```

Construct a new invalid action object.

Parameters

text	- A message text, describing the reason of this exception.
------	--

8.2.3 Member Function Documentation

8.2.3.1 what()

```
const char * untangle::invalid_action::what () const [inline], [override], [noexcept]
```

The message text describing the reason of this exception.

Returns

const char* - The message text.

The documentation for this struct was generated from the following file:

- actuator.hpp

Chapter 9

File Documentation

9.1 actuator.hpp

```
00001 // Copyright (c) 2018 Nicolae Popescu. MIT License.
00002
00006 #pragma once
00007
00008 #include <exception>
00009 #include <functional>
00010 #include <iostream>
00011 #include <list>
00012 #include <map>
00013 #include <memory>
00014 #include <string>
00015 #include <type_traits>
00016 #include <utility>
00017 #include <vector>
00018
00019 namespace untangle {
00020 // exception
00029 struct invalid_action : std::exception {
00035     explicit invalid_action(std::string text) : message(std::move(text)) {}
00036
00042     const char* what() const noexcept override { return message.c_str(); }
00043
00044 private:
00045     std::string message;
00046 };
00047
00055 template<typename... Args>
00056 auto last_arg(Args&&... args) {
00057     if constexpr (sizeof...(Args) <= 0) {
00058         return 0;
00059     } else {
00060         return std::get<sizeof...(Args)-1>(std::forward_as_tuple(args...));
00061     }
00062 }
00063
00072 template <typename action_t>
00073 struct actuator final {
00081     using actions_t = std::list<action_t*>;
00082     using actions_map_t = std::map<std::string, action_t*>;
00083     using result_t = std::conditional<std::is_void<typename action_t::result_type>::value, int,
00084                                     typename action_t::result_type>;
00091     using results_t = std::vector<typename result_t::type>;
00092
00093     actions_t actions;
00094     actions_map_t actions_map;
00095     results_t results;
00096
00097     actuator() = default;
00098     actuator(const actuator&) = default;
00099     actuator(actuator&&) noexcept = default;
00100     ~actuator() = default;
00107     actuator& operator=(const actuator& other) = default;
00108     actuator& operator=(actuator&&) noexcept = default;
00109
00117     action_t type() const { return nullptr; }
00118
00124     void reset() {
```

```

00125     actions.clear();
00126     actions_map.clear();
00127     results.clear();
00128 }
00129
00137 template <typename... Args>
00138 void operator()(Args&&... args) {
00139     results.clear();
00140
00141     // Dead bindings are collected here and dropped after the loop.
00142     // The actuator does not own the actions it points at, so it must never
00143     // write through action_t* into a std::function belonging to the caller.
00144     std::vector<action_t*> dead_actions;
00145
00146     for (const auto& action : actions) {
00147         // A null pointer or an empty std::function can never be invoked. Calling an
00148         // empty one throws std::bad_function_call, which is not an invalid_action and
00149         // would escape this operator, so drop it instead of invoking it.
00150         if (action == nullptr || !*action) {
00151             dead_actions.push_back(action);
00152             continue;
00153         }
00154
00155         try {
00156             if constexpr (std::is_same_v<typename action_t::result_type, void>) {
00157                 (*action)(std::forward<Args>(args)...);
00158             } else {
00159                 // invoke callback
00160                 auto last = last_arg(args...);
00161                 results.push_back((*action)(std::forward<Args>(args)...));
00162                 if constexpr (std::is_same_v<decltype(last), std::function<void(typename result_t::type)>») {
00163                     last(results.back());
00164                 }
00165             }
00166         } catch (const invalid_action& ia) {
00167             std::cout << ia.what() << std::endl;
00168             dead_actions.push_back(action);
00169         }
00170     }
00171
00172     for (const auto& dead_action : dead_actions) {
00173         actions.remove(dead_action);
00174     }
00175 }
00176
00183 template <typename... Args>
00184 void invoke_action(const std::string& name, Args&&... args) {
00185     results.clear();
00186     const auto& it = actions_map.find(name);
00187     if (it != actions_map.end()) {
00188         try {
00189             if constexpr (std::is_same_v<typename action_t::result_type, void>) {
00190                 (*it->second)(std::forward<Args>(args)...);
00191             } else {
00192                 results.push_back((*it->second)(std::forward<Args>(args)...));
00193             }
00194         } catch (const invalid_action& ia) {
00195             std::cout << ia.what() << std::endl;
00196             actions_map.erase(name);
00197         }
00198     }
00199 }
00200
00209 void add(action_t* action) { actions.push_back(action); }
00210
00217 void add(const std::string& name, action_t* action) { actions_map.emplace(name, action); }
00218
00229 void remove(const action_t* action) {
00230     actions.remove_if([&action](const auto& a) { return (action == a); });
00231 }
00232
00238 void remove(const std::string& name) { actions_map.erase(name); }
00239
00246 bool is_connected() const { return !actions.empty() || !actions_map.empty(); }
00247
00255 bool has_action(const std::string& name) const {
00256     return actions_map.find(name) != actions_map.end();
00257 }
00258 };
00259
00274 template <typename action_t, typename... Actions>
00275 actuator<action_t> connect(action_t& A1, Actions&&... An) {
00276     using actuator_t = untangle::actuator<action_t>;
00277     actuator_t actuator;
00278     actuator.actions = {&A1, &An...};
00279
00280     // remove empty actions

```

```

00281 actuator.actions.remove_if([](const auto& action) { return (*action == nullptr); });
00282 return actuator;
00283 }
00284
00285 template <typename actuator_t>
00286 void remove_empty_actions(actuator_t& actuator) {
00287     std::vector<typename actuator_t::actions_map_t::const_iterator> removable_iterators;
00288     for (auto it = actuator.actions_map.begin(); it != actuator.actions_map.end(); ++it) {
00289         if (it->second == nullptr) {
00290             removable_iterators.push_back(it);
00291         }
00292     }
00293     for (auto it : removable_iterators) {
00294         actuator.actions_map.erase(it);
00295     }
00296 }
00297
00298 template <typename key_t, typename action_t, typename... Actions>
00299 actuator<action_t> connect(std::pair<key_t, action_t*> A1, Actions... An) {
00300     using actuator_t = untangle::actuator<action_t>;
00301     actuator_t actuator;
00302     actuator.actions_map = {A1, An...};
00303
00304     // remove empty actions
00305     remove_empty_actions(actuator);
00306     return actuator;
00307 }
00308
00309 // generic helpers to remove const qualifier from a function type,
00310 // for instance const member functions
00311 template <typename T>
00312 struct function_remove_const;
00313
00314 template <typename R, typename... Args>
00315 struct function_remove_const<R(Args...)> {
00316     using type = R(Args...);
00317 };
00318
00319 template <typename R, typename... Args>
00320 struct function_remove_const<R(Args...) const> {
00321     using type = R(Args...);
00322 };
00323
00324 template <typename class_t, typename T,
00325         typename action_t = std::function<typename function_remove_const<T>::type>
00326         action_t bind(const std::shared_ptr<class_t>& obj, T class_t::* method) {
00327     return [wp = std::weak_ptr<class_t>(obj), method](auto&&... args) ->
00328         typename action_t::result_type {
00329         // lock() also keeps the object alive for the duration of the call
00330         const auto obj_ = wp.lock();
00331         if (obj_) {
00332             // inform the actuator about dead binding
00333             throw invalid_action("bind: invalid object");
00334         }
00335         return ((*obj_).*method)(std::forward<decltype(args)>(args)...);
00336     };
00337 }
00338
00339 template <typename class_t, typename T,
00340         typename action_t = std::function<typename function_remove_const<T>::type>
00341         action_t bind(class_t* obj, T class_t::* method) {
00342     return [obj, method](auto&&... args) -> typename action_t::result_type {
00343         if (obj) {
00344             // inform the actuator about dead binding
00345             throw invalid_action("bind: invalid object");
00346         }
00347         return ((obj)->*method)(std::forward<decltype(args)>(args)...);
00348     };
00349 }
00350
00351 // namespace untangle

```


Index

- actions__t
 - untangle::actuator< action__t >, [18](#)
- actuator, [1](#)
- add
 - untangle::actuator< action__t >, [20](#)
- bind
 - namespace untangle: functions, [12](#)
- connect
 - namespace untangle: functions, [11](#)
- has_action
 - untangle::actuator< action__t >, [22](#)
- invalid_action
 - untangle::invalid_action, [23](#)
- invoke_action
 - untangle::actuator< action__t >, [20](#)
- is_connected
 - untangle::actuator< action__t >, [21](#)
- last_arg
 - untangle, [15](#)
- namespace untangle: functions, [11](#)
 - bind, [12](#)
 - connect, [11](#)
- operator()
 - untangle::actuator< action__t >, [19](#)
- operator=
 - untangle::actuator< action__t >, [19](#)
- remove
 - untangle::actuator< action__t >, [21](#)
- reset
 - untangle::actuator< action__t >, [19](#)
- results__t
 - untangle::actuator< action__t >, [18](#)
- type
 - untangle::actuator< action__t >, [19](#)
- untangle, [15](#)
 - last_arg, [15](#)
- untangle::actuator< action__t >, [17](#)
 - actions__t, [18](#)
 - add, [20](#)
 - has_action, [22](#)
 - invoke_action, [20](#)
 - is_connected, [21](#)
 - operator(), [19](#)
 - operator=, [19](#)
 - remove, [21](#)
 - reset, [19](#)
 - results__t, [18](#)
 - type, [19](#)
- untangle::invalid_action, [22](#)
 - invalid_action, [23](#)
 - what, [23](#)
- what
 - untangle::invalid_action, [23](#)